3 **(a)** acceleration / force (directly) proportional to displacement	M1		
and *either* directed towards fixed point			
or acceleration & displacement in opposite directions	A1	[2]	
(b) **(i)** maximum / minimum height / 8 mm above cloth / 14 mm below cloth	B1	[1]	
(ii) **1.** $a = 11\,\text{mm}$	A1	[1]	
2. $\omega = 2\pi f$	C1		
$= 2\pi \times 4.5$			
$= 28.3\ \text{rad s}^{-1}$ (*do not allow 1 s.f.*)	A1	[2]	

(c) **(i)**	$v = \omega a$	C1	
	$= 28.3 \times 11 \times 10^{-3}$		
	$= 0.31\ \text{m s}^{-1}$ (*do not allow 1 s.f.*)	A1	[2]
(ii)	$v = \omega \sqrt{(a^2 - y^2)}$		
	$y = 3\ \text{mm}$	C1	
	$= 28.3 \times 10^{-3} \sqrt{(11^2 - 3^2)}$	C1	
	$= 0.30\ \text{m s}^{-1}$ (*allow 1 s.f.*)	A1	[3]

4 **(a)** $\Delta U = q + w$ (*allow correct word equation*) B1 [1]